Chapter 11
Designing Effective Output

Key Points and Objectives

1. The system analyst should design output to serve the intended purpose, be meaningful to the user, deliver the right quantity of output, deliver it to the right place, provide output on time, and choose the right output method.

2. Output can be in the form of a smart phone, tablet, print, display screen, audio, CD-ROM, DVD, email, faxes, or Web pages.

3. Output technologies differ in their accessibility, cost, durability, distribution, flexibility, transportability, and storage and retrieval possibilities.

4. Factors that the analyst must consider when choosing an output technology are:

   A. Who will use the output?
   B. How many people need the output?
   C. Where is the output needed?
   D. What is the purpose of the output? What user and organizational tasks are supported?
   E. What is the speed with which output is needed?
   F. How frequently will the output be accessed?
   G. How long will the output be stored?
   H. Under what special regulations is the output produced, stored, and distributed?
   I. What are the initial and ongoing costs of maintenance and supplies?
   J. What are the environmental requirements for output technologies?

5. Analysts must be aware of sources of output bias, interact with users to design output, inform users of the possibilities of bias in output, create flexible and modifiable output, and train users to use multiple output to help verify the accuracy of reports.

6. Bias is introduced in three main ways:

   A. How information is sorted
   B. Setting of acceptable limits
   C. Choice of graphics

7. Reports fall into three categories:

   A. Detailed
   B. Exception
   C. Summary

8. Guidelines for display design are:

   A. Keep the display simple
   B. Keep the presentation consistent

C. Facilitate user movement among displayed output
D. Create an attractive and pleasing display

9. When designing graphical output, the analyst must determine:

   A. The purpose of the graph
   B. The kind of data that need to be displayed
   C. The audience
   D. The effects on the audience of different kinds of graphical output

10. A dashboard can display a graph, a problem light, or text. An executive can find a dashboard to be extremely useful in making decisions, but only if the dashboard is designed properly.

11. Widgets and gadgets are small programs that reside either in a sidebar attached to a browser or program or even reside in a special layer on the desktop itself. They can be any type of program that may be useful to anyone interacting with a computer.

12. Design principles must be used when designing websites. These include:

   A. Using professional tools
   B. Studying other sites
   C. Using Web resources
   D. Examining the sites of professional website designers
   E. Using tools that you are familiar with
   F. Use storyboarding, wireframing, and mockups
   G. Consulting books
   H. Looking at examples of poorly designed pages
   I. Creating Web templates
   J. Using plug-ins, audio, and video sparingly

13. Storyboards are used when developing a website or app to show the differences between screens. It can show how a visitor to the site would navigate the website.

14. Wireframing allows the designer to plan the overall design, showing what element appears at each position on the page. It can also show the navigational design, showing how to move from one page to the next using buttons, tabs, links, and pull-down menus.

15. The term *wireframe* has largely been replaced with mockup, which show what the output and input will look like.

16. Planning a website involves:

   A. Designing the structure of the website
   B. Focusing on the content
   C. Using meaningful text
   D. Including appropriate graphics
   E. Using cascading style sheets (CSS)
   F. Using divisions and cascading style sheets or tables to enhance a layout
   G. Paying attention to the presentation of the website, with a consideration of download times
   H. Constructing navigational links

I. Promoting the website

17. It is important to include Web 2.0 technologies that focus on enabling and facilitating user-generated content and collaboration. These include:

    A. Blogs
    B. Wikis
    C. Links to social networks on which the company has a presence
    D. Tagging

18. Companies use collaborative tools to:

    A. Communicate an integrated branding and messaging strategy across multiple platforms
    B. To gauge consumer opinion
    C. To gather feedback
    D. To create a community of users

19. The following should be considered when designing for smartphones and tablets:

    A. Set up a developer account
    B. Choose a development process
    C. Be an original
    D. Determine how you will price the app
    E. Follow the rules for output design
    F. Design your icon
    G. Choose an appropriate name for the app
    H. Design for a variety of devices
    I. Design the output for the app
    J. Design the output a second time for different orientation
    K. Design the logic
    L. Create the user interface using gestures
    M. Protect your property
    N. Market your app

20. XML (extended markup language) may be transformed using cascading style sheets (CSS) or extensible style language transformations (XSLT) to create output.

21. Cascading style sheet output may be designed for different types of output media, such as print, Web pages, or handheld devices.

22. Ajax uses both JavaScript and XML to obtain small amounts of data from a server without leaving the Web page. The user does not have to wait for a new Web page to display after making a selection

Consulting Opportunity 11.1 (p. 298)

Your Cage or Mine

Some of the problems encountered by the committee regarding the output include the inconvenience of not having hard copy. The problem is increased by the fact that the computer is in Ella's office; it is therefore not so easily accessible to others. Summary reports of the zoo are not available, and decision makers do not receive information in time to make more effective decisions. Furthermore, the cost of reproducing financial data using the current method is too high. On top of that, only numbers can be output to hard copy, and printouts do not copy well.

To improve output for the committee, it should be designed to serve their needs more adequately. A summary of expenses, for example, should be supplied. It is also necessary to supply the information on time. The quantity of output needed is not handled by the current system. It is apparent that the current methods of output are not sufficient to meet the needs of the committee. The appropriate quantity of output must be delivered as well to solve the problem of distribution. Output must be more accessible to anyone in the committee who requires it. In conclusion, a system requirements analysis should be conducted to solve the problems experienced by the committee.

It may be a good idea to consider selling the old printer and buying a new one to replace it. The output technology in the zoo is out of date. That would help to justify the higher price. A final printer consideration is that the new printer will require less maintenance than the old one.

The Internet may be used in two ways. One is to use email bulk mailing lists to send information to other zoos when breeding situations arise, as well as communicate within committee members. A more important use of the Internet would be to create Web pages for communicating data on the animals and financial data. The Web pages could be linked to database information, which would be periodically updated.

Consulting Opportunity 11.2 (p. 301)

A Right Way, a Wrong Way, and a Subway

The output given to Rayl is too detailed; it does not serve the intended purpose nor is it designed to fit his use. What Rayl needed was a summary of ticket sales and the routes, not raw data. Strangely enough, the clerks in the booth can get summarized ticket sales onscreen whenever they want, yet the person who truly needs the information is unable to do so. The media being used for the output are not right. Although the timing of the distribution of information is good, it does not include all the necessary information required; therefore, a problem exists.

The external output for users of the computerized machines is not a good fit. It is too complicated for the users to understand.

Some of the changes in the output that need to be made are more timely distribution of output with the required information; and information in a more summarized form so that analysis can be made to help Rayl make decisions on fares and schedules.

The decisions that are being faced by collecting destination data is how to effectively store and use the data. The results of the data collection must be summarized in a way that will give a meaningful picture of subway usage, including which areas have heavy and lighter traffic. The use of smartphones to purchase tickets or for entrance may be considered.

Consulting Opportunity 11.3 (p. 303)

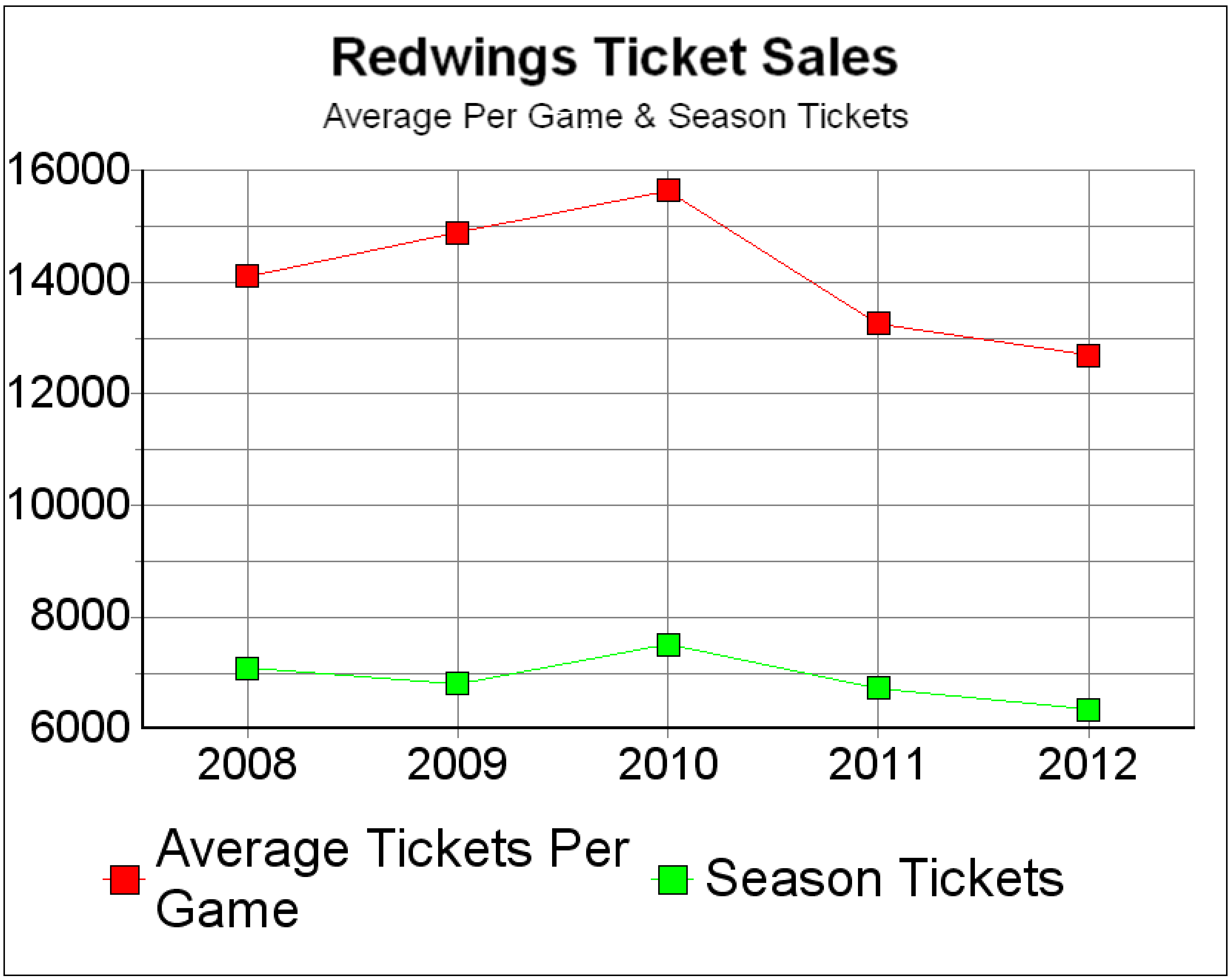
Should This Chart Be Barred?

Andy is talking about the trends of ticket sales over the years with a bar chart. However, a bar chart is never used to show a relationship over a period of years. Rather, Andy needs to prepare a line graph for the purpose. With a line graph, Andy may enjoy its strengths. That is, a line graph can show:

1. A peak point
2. A lowest point
3. Trends over the years

Redwings' ticket sales over the years can be represented as follows with a line graph, which is much more effective than a bar chart to analyze the trends over the years.

A suggested chart is illustrated below.



Consulting Opportunity 11.4 (p. 309)

Is Your Work a Grind?

There are several stylistic considerations that Paul ought to observe when redesigning the report. The output should be well organized, in the way that the eye sees. Related items, such as sales from the same store, or volume of items sold, should be grouped together. The report should also read from top to bottom and left to right.

Use of control breaks to set off a page of report would help create some aesthetic flavor. Control breaks can be used, for example, to draw attention to the summary of the number of pork items and volume of items sold in each outlet. Critical information can be highlighted with special characters such as asterisks to show any abnormal sales volume. The long, unbroken columns of information must be avoided. Too much cramming together causes headaches; additional blank spaces between columns actually contribute to readability of the report. Finally, the organization of the report is very important because there is so much information being presented.

The analyst could perhaps show an example of a report that is very crowded with information, and another that is neater and well formatted to suggest that the less-crowded format would ultimately be more beneficial. It is often a losing situation when a systems analyst must implement user suggestions that contradict his/her design training. Users will feel that their input is not appreciated if their suggestions are not taken; yet they are not satisfied when the final output is designed according to their misguided specifications.

The advantage to Stephen's large report is that he does not have to go through many pages of reports to find the information he wants. However, the tradeoff is that not all the information can be squeezed together in one report. Incorrect interpretations can occur, and bad decisions can be made; more often than not, it becomes confusing and difficult to get the entire picture of the report.

A good heuristic is to provide the user with many reports rather than one large report. Only in exceptional circumstances are long reports advantageous.

A Web-based solution would be useful as a replacement for much of the detailed information, because Stephen likes to receive an enormous amount of information. This could provide improved timeliness in addition to making information available to branch stores. A dashboard could be used to provide multiple outputs that would display up-to-date information in a variety of formats. However, summary reports are also required unless Stephen is planning on taking a laptop computer into meetings.

Consulting Opportunity 11.5 (p. 317)

A Field Day

Other guidelines to remember are to facilitate movement among displays and to create an attractive display format. The use of pull-down menus and button bars (with tool-tip help) that allow Fields to receive just the information he requires. Displays should be consolidated so that related, useful information is on each screen. Clear help must be easily available. Use of scroll bars and other GUI features would make the displays more useful.

A hyperlinked page should have choices available on the first page, which acts as a menu. Subsequent pages would contain detailed information, with links back to the main page as well as other related pages. The design of the hyperlinked pages will vary from student to student, but should contain the following:

- Display screens for listing the European markets. The display would have links for each European market.
- Display screens for the outlets to ship the goods to. Provide a link for each outlet.
- Display screens to decide how much merchandise to ship to each outlet. This could have links for each outlet, and provide information about inventory levels and sales at each store.
- Smartphone apps that include the above.

HyperCase Experience 11

1. Consider the reports from the training unit. What are Snowden's complaints about these reports? Explain in a paragraph.

   Model Solution: Not summarized, late, or not turned in at all.

2. Design a prototype output display based on the Training Unit's reports that will summarize the following information for Snowden.

   Number of accepted projects in the training unit
   Number of current projects currently being reevaluated
   Training subject areas for which consultation is being requested

   Model Solution: See the display screen illustrated below.

3. Display designs will vary from student to student. Students should keep in mind that Snowden Evans is the director of the Training and Management Systems Department and information should relate to these functional areas.

   Because Snowden already has in use a project status reporting system for Management Systems and the GEMS system makes heavy use of display screens, the new screens might relate to the reports already in use for the Training Unit.

   Some examples might be:

   Display screens to display information for training previously done for a client that is requesting additional training.

   A display screen to follow through on training leads from other departments.

   Display screens to summarize the progress of current Training Projects.

4. This is an activity that will vary from student to student.

5. The new display screens will depend on the feedback received and will vary from student to student.

Answers to Review Questions

1. *List six objectives the analyst pursues in designing system output.*

   Six objectives in designing system output include:

   A. Design output to serve a specific purpose
   B. Make output meaningful to the user
   C. Deliver the appropriate quantity of output
   D. Provide appropriate output distribution
   E. Provide the output on time
   F. Choose the most effective output method

2. *Contrast external outputs with internal outputs produced by the system. Remember to consider differences in external and internal users.*

   External output differs from internal output not only in its distribution, but in its design and appearance. Often external documents include instructions to the recipient and they include the company logo and corporate colors on printed or Web output. Instructions and the company logo may not be necessary on internal reports. The company logo and corporate colors may be included on internal websites.

3. *List potential electronic output methods for users.*

   Potential multimedia and electronic output methods are:

   A. FAX
   B. Email
   C. Web pages
   D. CD-ROM or DVD is also electronic output
   E. RSS feeds

4. *What are the drawbacks of electronic and Web-based output?*

   Drawbacks of electronic and Web-based output are:

   A. Special Web pages may need to be developed for users using handheld computers or mobile phones
   B. Displaying output on different screen resolutions
   C. Websites need diligent maintenance
   D. It is difficult to express a mood with electronic mail and communication may be more informal
   E. DVDs, CD-ROMs, and CD-RW require a computer for reading data

5. *List 10 factors that must be considered when choosing output technology.*

   Ten factors to consider when choosing output technology are:

   A. Who will use the output?
   B. How many people need the output?

C. Where is output needed?
D. What is the purpose of the output? What user and organizational tasks are supported?
E. What is the speed with which the output is needed?
F. How frequently will the output be needed?
G. How long will the output be stored?
H. Under what special regulations is the output produced, stored, and distributed?
I. What are the initial costs and ongoing costs of maintenance and supplies?
J. What are the human and environmental requirements for output technologies?

6. *What output type is best if frequent updates are a necessity?*

Display screen or Web output connected to online systems is best if frequent updates are a necessity.

7. *What kind of output is desirable if many readers will be reading, storing, and reviewing output over a period of years?*

If output is to be stored for long periods, a DVD or using the Web may be the best solution.

8. *What are two of the drawbacks to audio output?*

Two drawbacks of audio output are: (a) it is expensive to develop and (b) it needs earbuds where output will not interfere with other tasks.

9. *List three main ways in which presentations of output are unintentionally biased.*

Presentations of output can be biased by:

A. Sorting
B. Setting limits
C. Choice of graphics

10. *What are five ways the analyst can avoid biasing output?*

Strategies to avoid biasing output include:

A. Awareness of the sources of bias
B. Interactive design of output during prototypes that includes users and a variety of differently configured systems during the testing of Web document appearance
C. Working with users so that they are informed of the output's biases
D. Creating output that is flexible and allows users to modify limits and ranges
E. Training users to rely on multiple outputs for conducting "reality tests" on system output

11. *Why is it important to show users a prototype output report or display?*

Users should be shown a prototype of the output report so they can make changes. The workup should look as realistic as possible.

12. *List three categories of printed reports.*

The three categories of printed reports are summary, detailed, and exception reports.

13. Give one difference between exception and summary reports.

The difference between exception and summary reports are that summary reports contain information about all the data and exception reports contain information about data that meets certain conditions.

14. In what ways do displays, printed output, and Web-based documents differ?

Display output differs from printed output in that display output is ephemeral, can be more specifically targeted to the user, is available on a more flexible schedule, is not portable in the same way, can be changed through direct interaction, and requires that users be instructed on how to continue reading and how to end the display, user access may be through a password. Web-based output should include graphics, links, and other elements designed to draw the general user as well as make the viewer want to return to the site.

15. List four guidelines to facilitate the design of good display output.

Four guidelines to facilitate the design of good display output include:

A. Keep the display simple
B. Keep presentation consistent
C. Facilitate user movement among display output
D. Create an attractive and pleasing display

16. What differentiates output for a DSS from that of a more traditional MIS?

Output designed for decision support systems is characterized by greater modifiability and flexibility requirements than are traditional management information systems. All of the information they need to make decisions should be displayed in front of them and the data should be in the right form.

17. What are the four primary considerations the analyst has when designing graphical output for decision support systems?

Four primary considerations for designing graphical output for decision support systems include:

A. The purpose of the graph must be determined
B. The kind of data that needs to be displayed
C. Its audience
D. The effects on the audience of different kinds of graphical output

18. Define stickiness.

Stickiness is when a user stays at your website for a long period of time.

19. List seven guidelines for creating good websites.

Choose seven of the following ten guidelines for creating good websites:

A. Use professional tools

B. Study other websites
C. Use the resources that the Web has to offer, sites that give hints on design
D. Examine the websites of professional designers
E. Use tools that you have already learned
F. Consult books
G. Look at examples of poor websites
H. Create templates of your own
I. Use plug-ins, audio, and video sparingly
J. Plan ahead and think out the website
K. Use storyboarding, wireframing, and mockups

20. *List five guidelines for using graphics in designing websites.*

Choose five of the six guidelines listed below for using graphics in designing websites:

A. Use either JPEG, GIF, or PNG formats
B. Keep the background simple and readable
C. Create a few professional looking graphics for use on your page
D. Keep graphics images small and reuse images
E. Use the ALT attribute to display a small amount of text when the cursor moves over images and image hot spots
F. Examine your website on a variety of monitors and graphics resolutions

21. *List seven ideas for improving the presentation of corporate websites that you design.*

Choose seven of the following nine ideas to improve the presentation of a corporate website:

A. Provide an entry display or home page
B. Keep the number of graphics to a reasonable minimum
C. Use large and colorful fonts for headings
D. Use interesting images and buttons for links
E. Use cascading style sheets (CSS) to control the formatting and layout of the Web page
F. Use divisions and cascading styles or tables to enhance a layout
G. Use the same graphics image on several Web pages
H. Use JavaScript to enhance the Web page layout
I. Avoid overusing animation, sound, and other "busy" elements

22. *What is the "three-clicks" rule?*

A user should be able to move from the current page to the desired page in three clicks of a mouse button.

23. *In what ways can you encourage companies to promote their websites that you have developed?*

You can encourage companies to promote their websites by:

A. Including metatags used by search engines
B. Encouraging readers to bookmark your site (Internet Explorer "favorites")
C. Submitting your site to search engines
D. Purchase software to make the process of submitting your site to search engines easier

24. *What are Web 2.0 technologies? List four of the many in existence.*

Four of the many Web 2.0 technologies are blogs, wikis, links to social networks on which the company has a presence, and tagging.

25. *List the five factors analysts should consider when including Web 2.0 technologies in organizational web pages.*

The five factors analysts should consider when including Web 2.0 technologies in organizational web pages are:

A. Realize differences between corporate objectives and objectives of key stakeholders
B. Serve as the voice of the customer to your client organization
C. Recognize the importance of visual page design for effectively displaying collaborative tools
D. Revise and update the Web 2.0 technologies offered frequently
E. Work to integrate Web 2.0 technologies with the existing branding

26. *What is another word for app?*

Another word for app is application.

27. *List 7 of the 14 app development steps that deal with design.*

The list can include any of the following the 14 app development steps that deal with design:

A. Set up a developer account
B. Choose a development process
C. Be an original
D. Determine how you will price the app
E. Follow the rules for output design
F. Design your icon
G. Choose an appropriate name for the app
H. Design for a variety of device
I. Design the output for the app
J. 10. Design the output a second time for different orientation
K. Design the logic
L. Create the user interface using gestures
M. Protect your property
N. Market your app

28. *List two programs that can help you design output for a smartphone or tablet app.*

Two programs, selected from the following three that can help you design output for a smartphone or tablet app are AppCraftHD, iMockups, and App Cooker.

29. *What principles of agile development apply to developing an app?*

The principles of agile development apply to developing an app are quick releases and prototyping.

30. *Why is it important to design an app for different orientations (portrait and landscape)?*

    It is important to design an app for both portrait and landscape orientations because some apps look better in either landscape or in portrait mode, and when text is justified, the gaps between words seem to be larger in portrait mode, making it more difficult to read. Some apps, such as a calculator, change functionality when switched between landscape or in portrait mode.

31. *List the six basic options for pricing an app you develop.*

    The six basic options for pricing an app you develop are:

    A. Choose a low-cost strategy
    B. Introduce an app as a "premium" app
    C. Adopt a "freemium" model
    D. Offer an app for free
    E. Promote an app by reducing its price
    F. Accept advertising

32. *What are gestures in designing smartphones and tablets?*

    Gestures used when designing smartphones and tablets are swipes, pinches, tugs, and shakes.

33. *What are three ways to protect your app as intellectual property?*

    Three ways to protect your app as intellectual property are to trademark your icons and logos, copyright your app, and create your own end user license agreement (EULA).

34. *How do Cascading Style Sheets allow an analyst to produce output?*

    A cascading style sheet contains a series of styles such as font family, size, color, and so on, that are used to format a Web page or the elements of an XML document.

35. *What are the advantages of using XSLT instead of Cascading Style Sheets?*

    The advantages of using XSLT instead of a cascading style sheet are that they allow the analyst to select the elements, sort the elements, and transform the XML document into a variety of output media.

36. *What are RSS feeds?*

    RSS (really simple syndication) feeds are XML documents that users can obtain from links on Web pages or to which they can subscribe.

37. *What are dashboards mainly used for?*

    Dashboards are used for communicating measurements to the user. Executives use a dashboard to review performance measures and, if the screen calls for it, to take action on the information.

38. *What are widgets (or gadgets)?*

    Widgets and gadgets are used to personalize desktops. They are small programs that reside either in a sidebar attached to a browser or program or even reside in a special layer on the desktop itself.

39. *Why should a systems designer be aware of the popularity of widgets (or gadgets)?*

Designers can learn a lot about what users prefer when they study user-designed desktops. Because widgets and gadgets can also distract people from system-supported tasks, designers need to work with users to support them in achieving a balance.

40. *How does Cascading Style Sheets allow an analyst to produce output?*

Cascading style sheets are commonly stored in a file external to the Web page, and one style sheet may control the formatting of many pages. Changing the external style sheet will change the formatting of all the Web pages that use the style sheet. Divisions combined with cascading styles control the layout by providing blocks of text on the Web page, and are better suited for screen reading software for visually impaired site visitors.

A style sheet provides a series of styles, such as font family, size, color, border, and so on, that are linked to the elements of an XML document. These styles may vary for different media, such as a screen, printed output, or a handheld device.

41. *What are the advantages of using Extensible Style Sheet Language Transformations instead of Cascading Style Sheets?*

Extensible style language transformations (XSLT) allow the analyst to select elements, sort elements, and insert them into a Web page or another output medium.

42. *How does Ajax help to build effective Web pages?*

Ajax uses both JavaScript and XML to obtain small amounts of data, either plain text or XML, from a server without leaving the Web page. The entire Web page does not need to be reloaded, but is instead reformatted based on choices that a user inputs.

Problems

1. *"I'm sure they won't mind if we start sending them the report on these oversized computer sheets. All this time we've been condensing it, retyping it, and sending it to our biggest accounts, but we just can't now. We're so understaffed, we don't have the time," says Otto Breth. "I'll just write a comment here telling them how to respond to this report, and then we can send it out."*
   a. *What potential problems do you see in casually changing external output? List them.*
   b. *Discuss in a paragraph how internal and external output can differ in appearance and function.*

   a. Some of the potential problems that can be caused by changing external outputs casually are:

   Initial confusion
   Dissatisfaction
   Resistance

b. External outputs usually include instructions to the recipients. Some are placed on preprinted forms bearing the company logo, address, and corporate colors. Internal outputs range from short summary reports to lengthy, detailed reports to help decision makers make more effective decisions. Attractive design and appearance are usually of greater importance to external users.

2. *"I don't need to see it very often, but when I do, I have to be able to get at it quickly. I think we lost the last contract because the information I needed was buried in a stack of paper on someone's desk somewhere," says Luke Alover, an architect describing the company's problems to one of the analysts assigned to the new systems project. "What I need is instant information about how much a building of that square footage cost the last time we bid it; what the basic materials such as steel, glass, and concrete now cost from our three top suppliers; who our likely competition on this type of building might be; and who comprises the committee that will be making the final decision on who gets the bid. Right now, though, it's in a hundred reports somewhere. I have to look all over for it."*
   a. Given the limited details you have here, write a paragraph to suggest an output method for Luke's use that will solve some of his current problems. In a second paragraph, explain your reasons for choosing the output method you did. (Hint: be sure to relate output method to output content in your answer.)
   b. Luke's current thinking is that no paper record of the output discussed needs to be kept. In a paragraph, discuss what factors should be weighed before displayed output is used to the exclusion of printed reports.
   c. Make a list of five to seven questions concerning the output's function in the organization that you would ask Luke and others before deciding to do away with any printed reports currently being used.

a. Display output should be used as the output method for Luke's system. Printed output should also be available to print the information from the display screen for hard copy. Display output is more flexible, allowing Luke to make changes, retrieve previous reports on the bids, and make decisions instantaneously. Some output could be on tablets or smartphones.

b. Factors that should be weighed before display output is used in exclusion of printed output are:

Transportability
Ability to retrieve information about previous bids

c. The following questions will help to determine if printed output is necessary:

Where are the deals closed?
How does the output function as part of the bid procedure?
Is a computerized facility available during the bidding?
Is printed output required to be submitted for consideration by the committee?
Is printed output required for past bids?
Would it make any difference if the printed output were redesigned to fit more information in fewer reports?

3. *Here are several situations calling for decisions about output content, output methodology, distribution, and so on. For each situation, note the appropriate output decision.*

a. A large, well-regarded supplier of key raw materials to your company's production process requires a year-end summary report of totals purchased from it.
b. Internal brainstorming memos are circulated through the staff regarding plans for a company picnic and fundraiser.
c. A summary report of the company's financial situation is needed by a key decision maker, who will use it when presenting a proposal to potential external backers.
d. A listing of the current night's hotel room reservations is needed for front desk personnel.
e. A listing of the current night's hotel room reservations is needed by the local police.
f. A real-time count of people passing through the gates of Wallaby World (an Australian theme park) will be used by parking lot patrols.
g. An inventory system must register an item each time it has been scanned by a wand.
h. A summary report of merit pay increases allotted to each of 120 employees will be used by 22 supervisors during a joint supervisors' meeting, and subsequently when explaining merit pay increases to the supervisors' own departmental employees.
i. Competitive information is needed by three strategic planners in the organization, but it is industrially sensitive if widely distributed.
j. A casual style of conversation is needed to inform customers about powerful but seldom used features of a product.
k. A historic district of a city wants to let visitors know about historical buildings and events.
l. Storm warnings must be delivered to subscribers in a large geographical area.

a. Printed output
b. Display output, Web page, smartphone
c. Display output
d. Display output
e. Display output
f. Smartphone or display output
g. Display output
h. Display and printed output
i. Display output
j. Corporate blog, smartphone, or tablet output
k. Podcast, smartphone output
l. Smartphone, tablet, and widgets

4. "I think I see now where that guy was coming from, but he had me going for a minute there," says Miss deLimit. She is discussing a prototype of display output, one designed by the systems analyst that she has just seen. "I mean, I never considered it a problem before if even as much as 20 percent of the total class size couldn't be fit into a class," she says. "We know our classes are in demand, and because we can't hire more faculty to cover the areas we need, the adjustment has to come in the student demand. He's got it highlighted as a problem if only 5 percent of the students who want a class can't get in, but that's okay. Now that I know what he means, I'll just ignore it when the computer beeps."

a. In a sentence or two, describe the problem Miss deLimit is experiencing with the display output.
b. Is her solution to "ignore the beeps" a reasonable one given that output is in the prototype stage?
c. In a paragraph, explain how the display output for this particular problem can be changed so that it better reflects the rules of the system Miss deLimit is using.

a. Miss deLimit is experiencing the bias set by the low limit of the systems analyst regarding class demand.

b. Because output is in the prototype stage, she should let the system analyst know about her reaction so that he can make the necessary changes, instead of ignoring the "beeps." She is not contributing to the prototype if there is no feedback.

c. The display output should allow the user to change limits used in the display screen to reflect rules of the system.

5. *Following is a log sheet for a patient information system used by nurses at a convalescent home to record patient visitors and activities during their shifts. Design a printed report using form design software that provides a summary for the charge nurse of each shift and a report for the activities coordinator at the end of a week. Be sure to use proper conventions to indicate constant data, variable data, and so on. These reports will be used to determine staffing patterns and future activities offerings.*

Date Z9-99-99		Daily Log of Patient Visitors & Activities	Page 99
Patient Name	No. Of Visitors	Relation	Activities
X-----------------X	99	X-------X	X---X
		X-------X	X---X
		X-------X	X---X
		X-------X	X---X
X-----------------X	99	X-------X	X---X
		X-------X	X---X
		X-------X	X---X
Total No. Of Visitors: 999			

6. *Design display output for Problem 5 using form design software. Make any assumptions about system capability necessary and follow display design conventions for onscreen instructions. (Hint: you can use more than one display screen if you wish.)*

   a. *In a paragraph, discuss why you designed each report as you did in Problem 5. What are the major differences in your approach to each one? Can the printed reports be successfully transplanted to displays without changes? Why or why not?*

   b. *Some of the nurses are interested in a Web-based system that patients' families can access from home with a password. Design an output screen for the Web. In a paragraph, describe how your report had to be altered so that it could be viewed by one patient's family.*

```
Z9-99-99                    Patient Visitors & Activities Log                    Z9:99

Enter Date (MM/DD/YY):  99/99/99

Patient Name: X----------------------------------------------------X (Last, First)

         Relation               Y/N        Count
         --------               -----      -------

         Father :                 X
         Mother :                 X
         Spouse :                 X
         Brothers & Sisters :     X          99
         Friends :                X          99
         Coworkers :              X          99
         Other :                  X          99

ACTIVITIES:   X----------------------------------------------------------------------------X
              X----------------------------------------------------------------------------X
              X----------------------------------------------------------------------------X
```

a. The printed report is designed landscape orientation. A date entry is not required. The display design requires scrolling, the equivalent to flipping pages, and a date entry to select the report date. Further, the display screen entry may have more information, or information presented in a more lengthy manner, because paper costs are not a consideration.

b. The report would have to be modified to include an entry field for a patient number and password, images for navigation (next, back, home), it would include only the patient information (not other patients), and a link to a privacy statement should be added to the Web page.

7. *Clancy Corporation manufactures uniforms for police departments worldwide. Its uniforms are chosen by many groups because of their low cost and simple but dignified design. You are helping to design a DSS for Clancy Corporation, and it has asked for tabular output that will help it in making various decisions about what designers to use, where to market its uniforms, and what changes to make to uniforms to keep them looking up to date. The following table lists some of the data the company would like to see in tables, including uniform style names, an example of a buyer group for each style, and which designers design which uniform styles. Prepare an example of tabular output for display that incorporates these data about Clancy's. Follow proper conventions for tabular output displays. Use codes and a key where appropriate.*

9/99/99 Clancy Corporation Z9:99

Uniform Purchases Summary

StyleBuyersDesigners

X--------------------X X-------------------------X X------------------------------X
X--------------------X X-------------------------X X------------------------------X
X--------------------X X-------------------------X X------------------------------X

8. *Clancy's is also interested in graphical output for its DSS. It wants to see a graphical comparison of how many of each style of uniform are being sold each year.*

   a. *Choose an appropriate graph style and design a graph for display that incorporates the following data:*

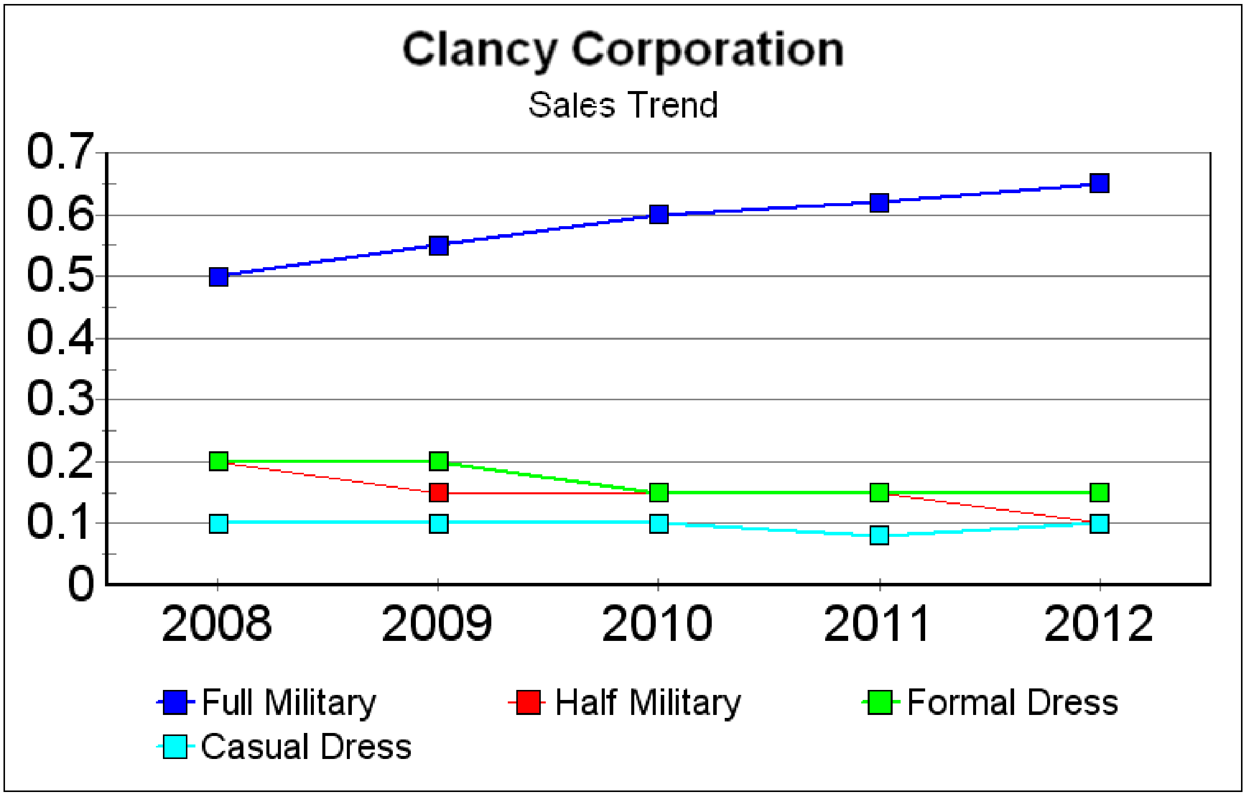
	Full Military (percent of total)	**Half Military**	**Formal Dress**	**Casual Dress**
2008	*50*	*20*	*20*	*10*
2009	*55*	*15*	*20*	*10*
2010	*60*	*15*	*15*	*10*
2011	*62*	*15*	*15*	*8*
2012	*65*	*10*	*15*	*10*

   Be sure to follow proper design conventions for displays. Use codes and a key if necessary.

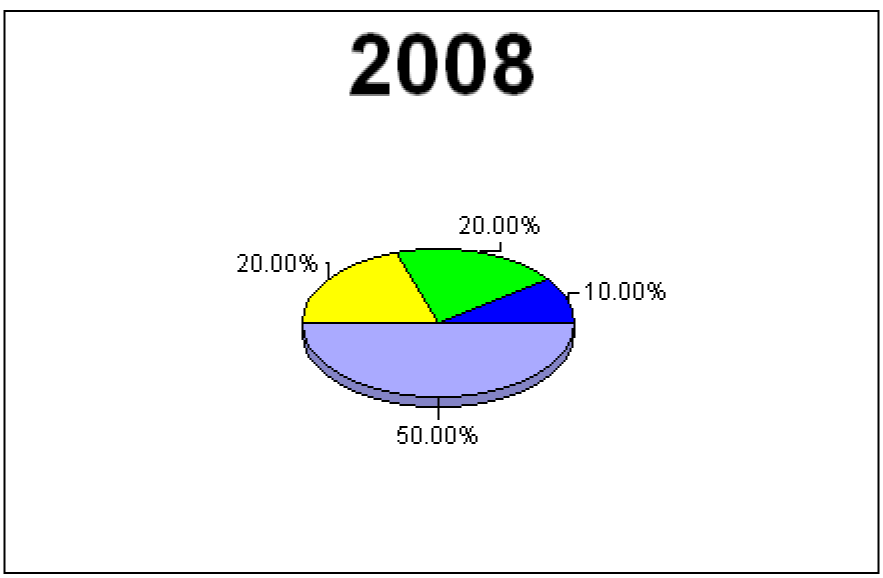
   b. *Choose a second method of graphing that might allow the decision makers at Clancy's to see a trend in the purchase of particular uniform styles over time. Draw a graph for display as part of the output for Clancy's DSS. Be sure to follow proper design conventions for displays. Use codes and a key if necessary.*

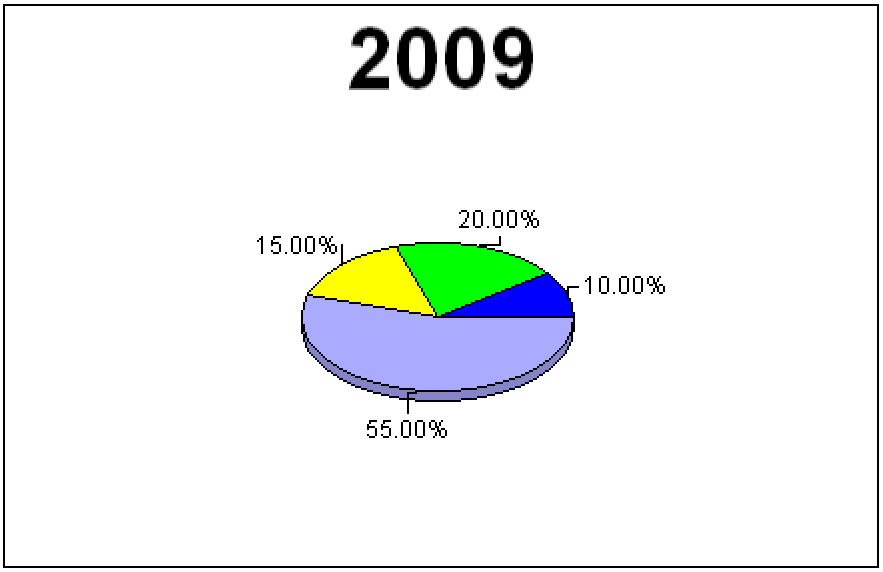
   c. *In a paragraph, discuss the differences in the two onscreen graphs you have chosen. Defend your choices.*

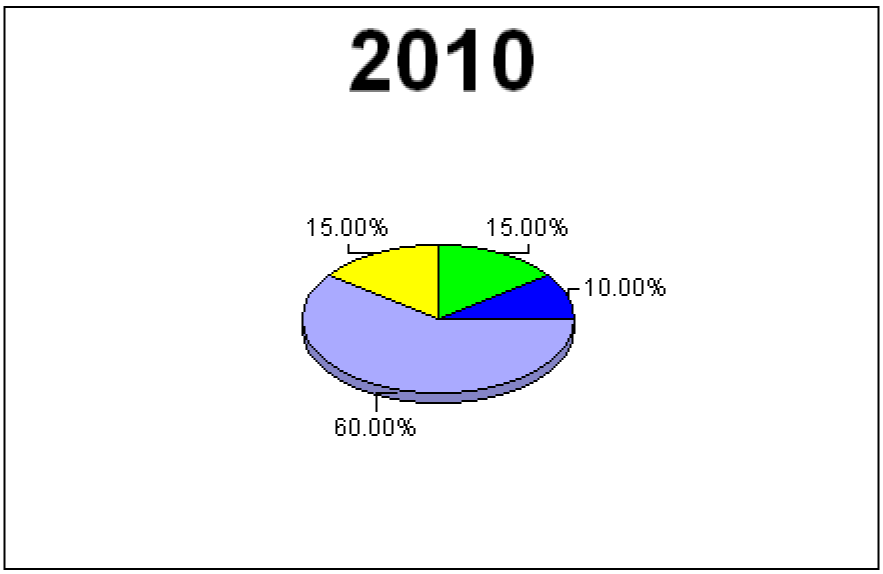
a.

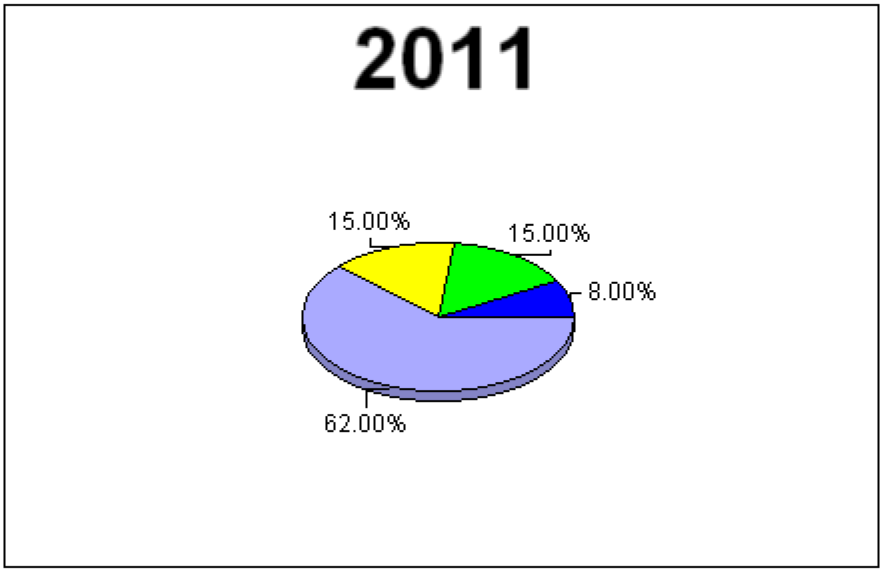

Clancy Corporation
Sales Trend
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
2008
2009
2010
2011
2012
Full Military
Half Military
Formal Dress
Casual Dress

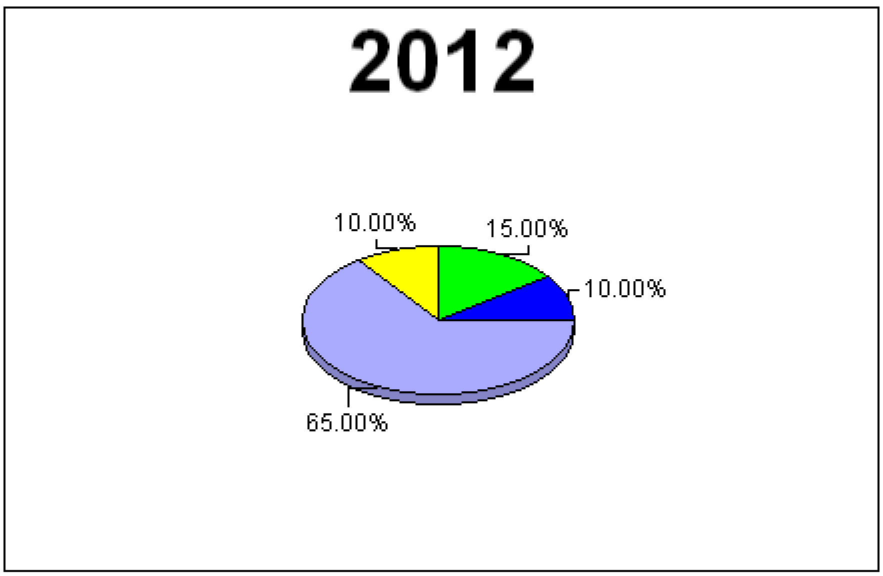
b.


2008
20.00%
20.00%
10.00%
50.00%


2009
15.00%
20.00%
10.00%
55.00%


2010
15.00%
15.00%
10.00%
60.00%





c. Pie charts are good for showing parts of a whole, whereas line graphs are appropriate for trends. In this case the line graph compares all years whereas the pie chart shows the parts just for one year.

9. *Michael Cerveris owns a number of cars used for racing. What performance measures does he need to develop to keep track of the performance of his driver, pit crews, and support staff (not to mention any bald tires his cars experience)?*

Michael should use a dashboard showing charts of his driver times, pit crew times, and support staff information. Line graphs and bar charts may give a picture of how much time it takes to change tires and so on.

10. *Design a DSS dashboard for Michael (Problem 9). Use appropriate types of charts and graphs to illustrate performance.*

    The design will vary from student to student. The best type of chart would be a line graph or a bar chart. The line graph would plot time to perform an action or complete a race on the Y axis and the different races on the X axis. It would provide a long range view of the data.

11. *Design a dashboard for keeping track of a person's stock and portfolio. Think about how the dashboard could be used to make decisions about buying and selling stock. Remember that a client can have more than one stockbroker.*

    The dashboard design will vary from student to student, but should include a chart of how each stock is doing over time. There might be one chart with different lines on it or a chart for each stockbroker, perhaps on a different tab. It could also have current values for each stock in a table. It might use a meter, similar to a speedometer, to show if the stock is above its recent average or below it.

12. *Gabriel Shanks runs a nonprofit theatre that produces seven plays per year in three theatres. Each play lasts eight weeks but can be extended four weeks if the show is a success. Design a dashboard for Gabriel, taking into consideration the different phases of putting on a performance as well as the need to sell as many tickets as possible. Don't forget that Gabriel is involved in theatre and is very visual. He doesn't like tables, however.*

    The design will vary from student to student. Because Gabriel is very visual, the dashboard should use charts and graphs to show ticket sales. It could use meters, similar to an automobile speedometer, to show above and below average ticket sales for each play. If the graphs show that sales are very high, an indicator light could turn green. If the play is not doing well, the indicator light could turn red.

13. *While Gabriel (from the previous problem) is taking care of various details during an ordinary day, he would like to keep up on theatre news in Manhattan, at the same time having some simple tools around to help him with his computer related activities. What sort of widgets and gadgets would Gabriel need to do his job while having some simple computer-based tools always available?*

    Gabriel could use a widget that would provide a news feed for the theatre news in Manhattan. Other computer-related widgets that he could use might be for Web searching, a people finder, perhaps one for travel search, and a weather widget. Students may come up with many more examples, and the number of widgets available is growing at a fast pace.

14. *Browse the Web to view well-designed and poorly designed websites. Comment on what makes the sites good or bad, using the critique form presented earlier in the chapter to compare and contrast them.*

    The results of this exercise will vary, depending on the website that students have reviewed.

15. *Propose a website for Clancy's, the uniform company described in Problems 7 and 8. Sketch by hand or use form design software to create a prototype of a Clancy's home page. Indicate hyperlinks and include a sketch of one hyperlink document. Remember to include graphics, icons, and even sound or other media if appropriate. In a paragraph, describe who the intended users of the website are and state why it makes sense for Clancy's to have a Web presence.*

The website will vary by a wide margin, depending on the students. Suggestions are:

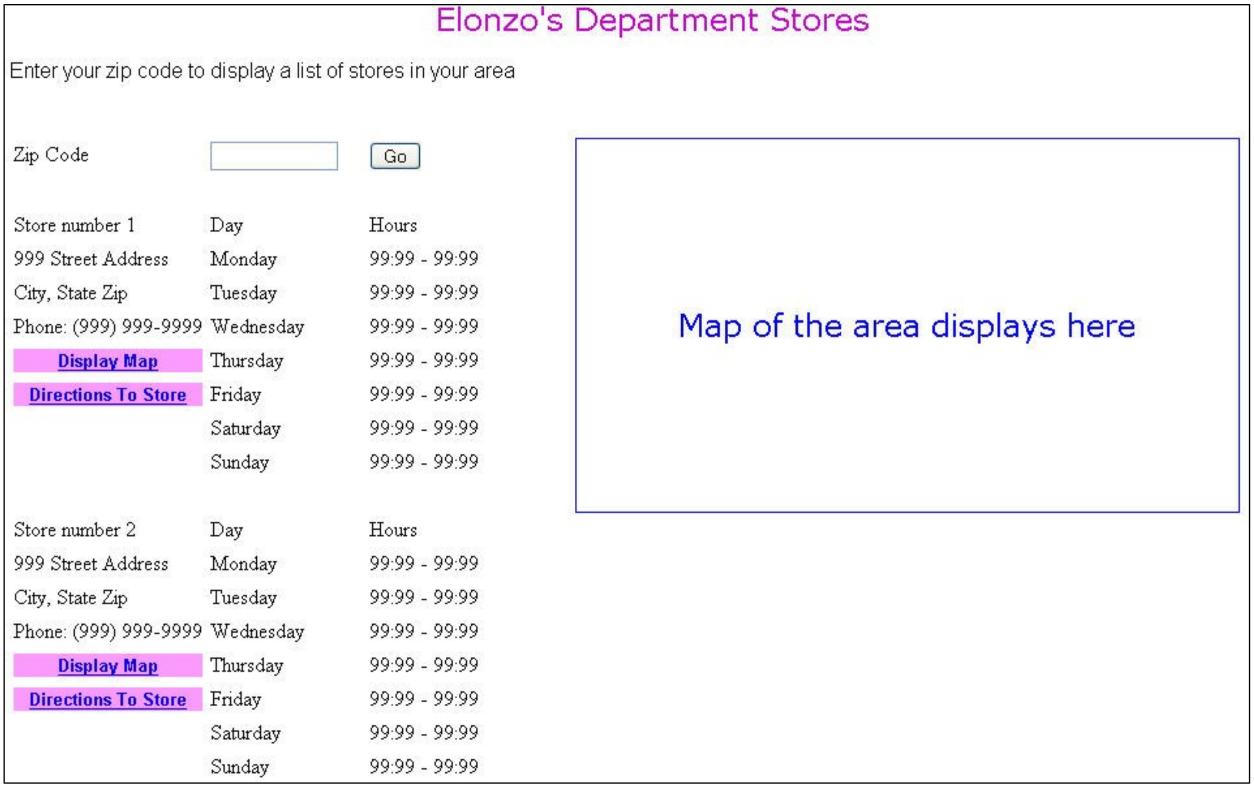
A. Put in graphics, perhaps a uniform or police badge.
B. The title should be in large letters.
C. There should be corporate information on the home page.
D. Links should be provided for each uniform style.
E. Each different style should have its own page, with an image or two on the styles.
F. Prices may be included on each page.
G. There might be links to the designers of the uniforms.
H. There should be an order form on a Web page. Each page should have a link to the order form.

The intended users of the website would be various police departments around the world. It makes sense to have a Web presence because Clancy is selling to the police departments, and there are many smaller departments that are potential customers.

16. *Elonzo's Department Stores is a chain of about 50 retail stores, specializing in kitchen, bath, and other household items, including many decorative and fashionable items. Recently Elonzo's decided to automate its gift registry to allow wedding and other event guests to be able to browse for items that were selected by the wedding couple or others.*
   a. *Design a Web page that would allow customers to enter a zip code and find the nearest store.*
   b. *Design a Web page for customers to browse gifts and order them online. Do not include the actual ordering forms, simply the products. What sort of options should be available for customers? Include buttons or links to change the sort sequence in your design.*
   c. *Design a printed list that customers could request when they go to one of the stores. What sequences would be optimal for a customer trying to find items? Would all items requested by the wedding couple be included on the list? (Hint: some may have been purchased already.)*

   a. All the designs will vary from student to student. Some examples are illustrated below. These do not include images or other graphics that students may design and include on the Web page.

The store locator might include a map or driving directions to the store. A sample Web page is included below.



b. A suggested website is included below. Student work may vary according to font characteristics, images, and content based on their own experience with gift registries. The information about gifts should be sorted by price, category, and description.

Elonzo's Department Stores
Gift Registry

Gift Registry for ----- bride and groom or other name here -----

Click to change the sort order [Price] [Category] [Description]

Product Description	Price	Category	Product Code
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX

c. The sort sequences should be by price, location of the item in the store, and product description. The gift list should only include items that have not already been purchased by people for the couple or person that has registered. An example of the printed report follows. This includes a bar code and registration number. When a gift is purchased, the bar code would be scanned and the product code would be scanned. This would transmit information to the central database so that all stores would know that the selected product has already been purchased.

Elonzo's Department Stores
Gift List

Gift Registry for ----- bride and groom or other name here -----

Product Description	Price	Store Location	Category	Product Code
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXX	ZZZ9.99	XXXXXXXXXXXXXXX	XXXXXXXXXXXXXXXXX	XXXXXXXXX

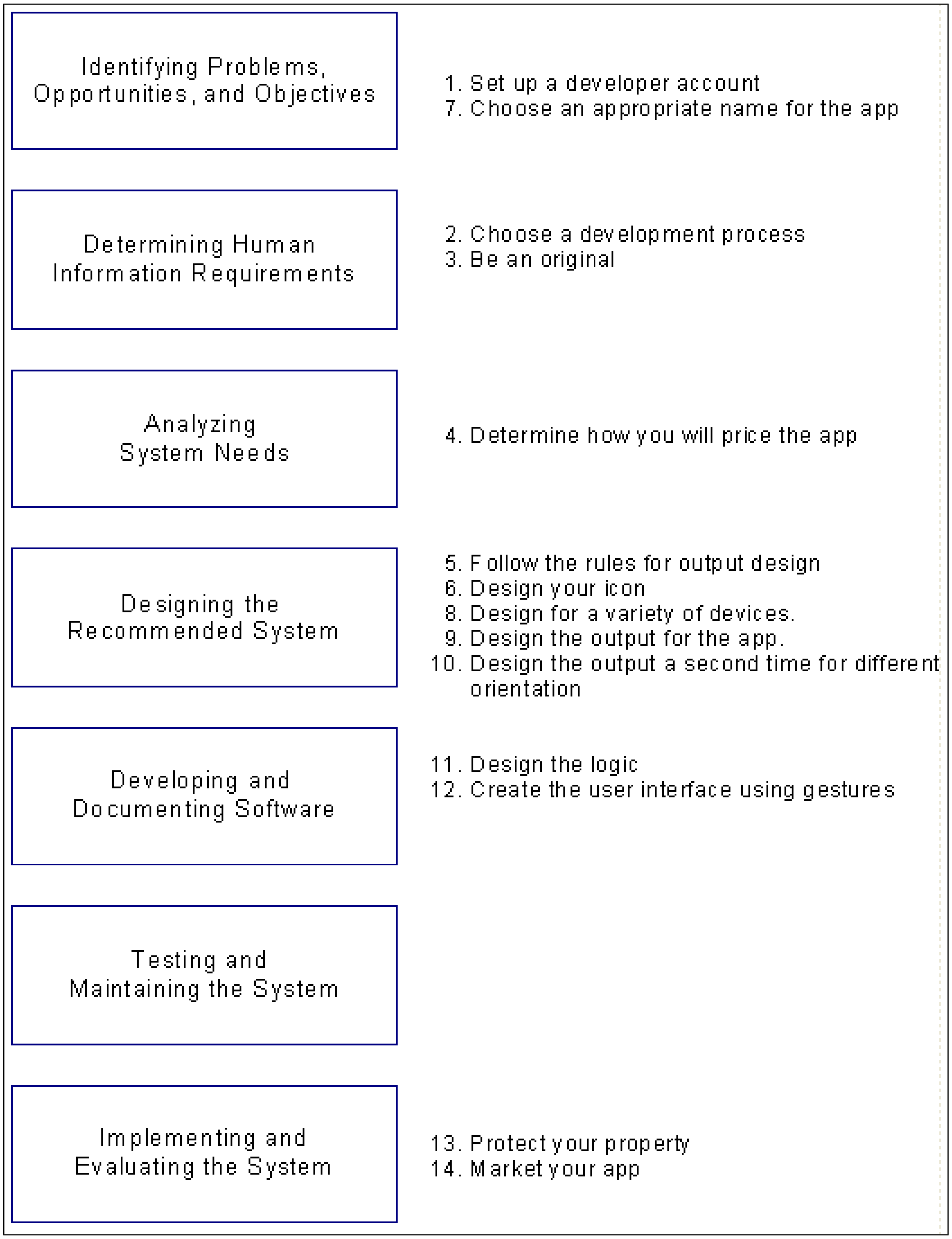
Registration number: 9999999

17. *Design an outline of a podcast for someone touring your university, college, or business. What sequence would you place the topics in? How much time would you allow for each campus or building location? Assume the party will arrive in the morning and sequence lunch into the podcast.*

    The design will vary from student to student, depending on campus or business size. It should have a logical sequence that would start at one location and move to an adjacent location. The tour should start at the reception desk or welcome center. It should be sequenced so that the visitor would arrive at the student union, cafeteria, or other public dining facility at lunch time. Each segment should include interesting information about the building, artifact, or other feature, and should be scripted so that the visitor does not backtrack. The final location will be back at the starting point.

18. *Draw a diagram that compares the flow of app development through the 14 steps with the seven phases of the SDLC.*

    A suggested diagram follows. This will vary depending on how the student creates the actual diagram and divides up the phases.



19. *Write two paragraphs that compare and contrast the process of app development and the SDLC as you diagrammed it in Problem 18.*

There are some significant differences between app development and the phases of the SDLC. The first is that there are no current problems, just opportunities. There is also no modeling of the current system, just the new system. There are no presentations to users and approvals.

20. *Design an airline flight reminder screen for a smartphone using portrait orientation.*

The design should be a simple text message with perhaps a graphic image above the message.

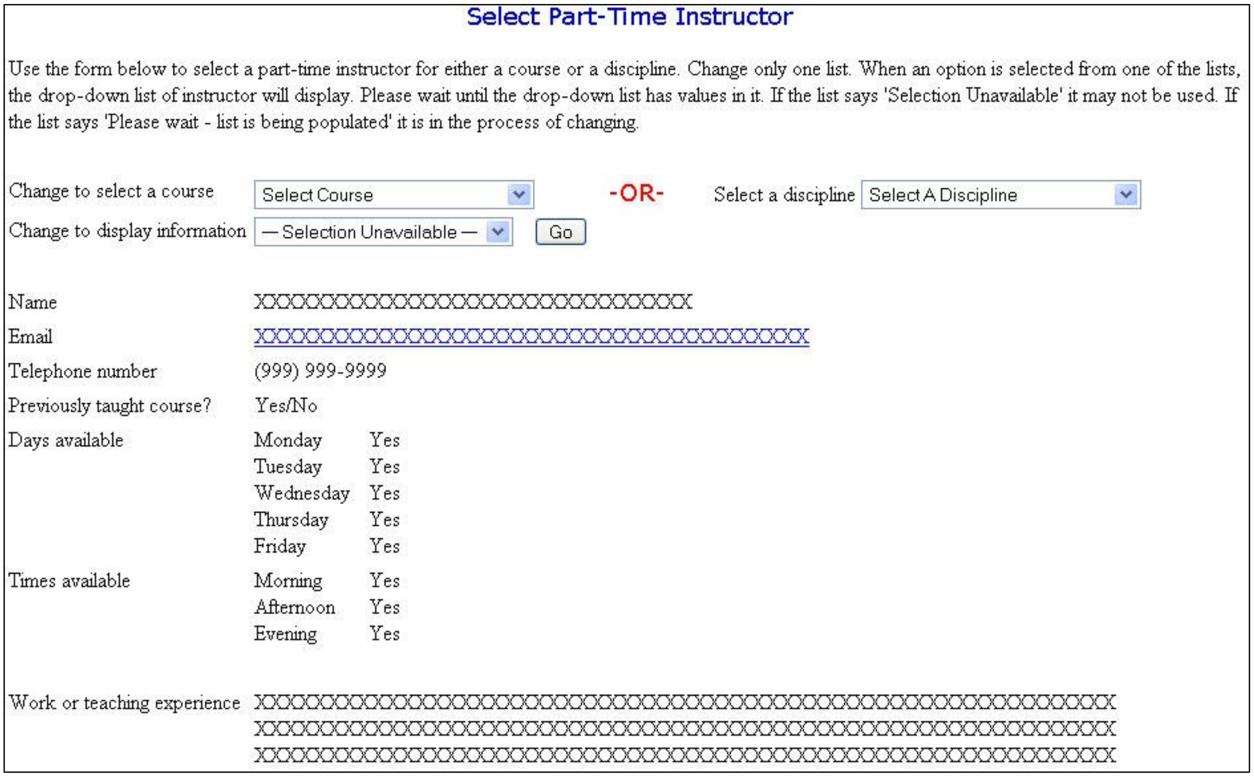
The design should include the time that the reminder was sent, along with the flight number, and whether the flight is on time or late and the flight status. It should allow you to check in using your app. Further features would include an airport map showing the terminal and the gate location (if it is available). If there is an airline lounge, it should also be included. A link would list your reservations. Contact information should be provided. An advanced feature would be an electronic boarding pass.

21. Design an airline flight reminder screen for tablet using landscape orientation.

Use the same considerations as above, but with the corporate icon on the left and the information on the right. It would also include a larger summary of the current itinerary, with maps and a flight map. It might have a link to your frequent flier account and a place for you to make notes. An advanced feature would be the ability to monitor your status on a standby list, and would allow you to make future reservations. If the flight is canceled or you miss your flight, a list of the next available flights should display.

22. *Design an Ajax style of Web page that would allow a dean at a community college to select part-time instructors. The dean should be able to select a discipline or a course and have the server send an XML document containing all the potential part-time instructors for the selection. The XML document should be used to populate a drop-down list of the instructor names. Clicking an instructor's name would display information about the potential instructor. Decide what information to include that would help the dean make a decision on whom to hire. (Hint: part-time instructors may be able to teach only on certain days or only in the morning, afternoon, or evening.)*

A suggested Web page is illustrated below.


Select Part-Time Instructor

Use the form below to select a part-time instructor for either a course or a discipline. Change only one list. When an option is selected from one of the lists, the drop-down list of instructor will display. Please wait until the drop-down list has values in it. If the list says 'Selection Unavailable' it may not be used. If the list says 'Please wait - list is being populated' it is in the process of changing.

Change to select a course | Select Course | -OR- | Select a discipline | Select A Discipline

Change to display information | — Selection Unavailable — | Go

Name	XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX	
Email	XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX	
Telephone number	(999) 999-9999	
Previously taught course?	Yes/No	
Days available	Monday	Yes
	Tuesday	Yes
	Wednesday	Yes
	Thursday	Yes
	Friday	Yes
Times available	Morning	Yes
	Afternoon	Yes
	Evening	Yes
Work or teaching experience	XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX	

Group Projects

1. *Brainstorm with your team members about what types of output are most appropriate for a variety of employees of Dizzyland, a large theme park in Florida. Include a list of environments or decision-making situations and types of output. In a paragraph, discuss why the group suggested particular options for output.*

Output suggestions are (students may come up with many more):

Human resources

Character manager: Character appearances at park locations display screen

Refreshment carts: Persons assigned to refreshment carts at various time slots

Maintenance crews: Cleaning crews assigned to various areas of the park

Equipment maintenance and repair information
Periodic maintenance of ride equipment

Sales and marketing

Ticket sale information for daily visitors
Hotel room reservation information
Ride statistics
Injuries and other problems

2. *Have each group member design an output display or form for the output situations you listed in Project 1. (Use either Microsoft Visio, a CASE tool or paper layout form to complete each display or form.)*

The display or form design contents will vary from group to group.

3. *Create a dashboard for Dizzyland in Group Project 1.*

The dashboard will vary depending on what student groups choose to focus on, but may include ticket sales in the form of a meter, hotel occupancy, charts of ticket sales and hotel occupancy, and so on.

4. *Design a website, either on paper or using software with which you are familiar, for Dizzyland. Although you may sketch documents or graphics for hyperlinks on paper, create a prototype home page for Maverick, indicating hyperlinks where appropriate. Obtain feedback from other groups in your class and modify your design accordingly. In a paragraph, discuss how designing a website is different from designing displays for other online systems.*

The website will vary from group to group but in general should include:

A. Images, colorful, showing people and families having fun, cartoon characters welcoming the visitor
B. The title should be in large, colorful letters
C. There should be corporate information on the home page
D. Links should be provided for different areas of the theme park, hotels, restaurants, and other main features
E. Maps of the park
F. Ticket prices and Web specials
G. Purchase tickets and reserve a room
H. There should be a contact us or feedback form on a Web page

5. *Without looking at your phone or tablet, design three sample pages for the following apps. You do not need to use a mock-up app for this assignment. Just draw them on paper. (Note that each group may be assigned a different app to do).*
a. *A grocery list helper*
b. *A hotel and room finder*
c. *An app to help you get to sleep*
d. *A task management (to-do) list*
e. *An apartment finder*
f. *A weather app*

The final mockups on paper will vary depending on the student group. Some things to look for are listed below. Use a tablet or smartphone to go to the app store or market place and find apps and compare them to the student results.

a. A grocery list helper

Include icons for categories and items. Select categories and then select items in each category using check boxes. Enter amounts needed. The app could also include a search area and entering the first few letters would bring up matches. It would also allow you to enter new items, such as ethnic food items, that would not be on the list. When the list is complete, display a shopping list sorted by category. Some recipe sites will allow you to create a shopping list from the selected menus.

An advanced feature would allow you to select a store and display a store map with items highlighted.

b. A hotel and room finder

The hotel finder app would use the current location or select a location (since the person would be traveling to that location) and display either all hotels or select hotels by brand for those enrolled in loyalty programs. The app would allow you to select by price or by amenities, such as a swimming pool or fitness room, free breakfast, etc. It should also show hotels on a map. The result list should show price and availability. Special sales and last minute deals should display.

An advanced feature would be using the app to make the hotel reservation.

c. An app to help you get to sleep

A sleep app could generate relaxing sounds or white noise to help you sleep. It could include guided meditations and other relaxation exercises, as well as tips for having good sleep.

d. A task management (to-do) list

A task management app should allow you select calendar days and times and enter the activities for those dates and times. People involved with the activity should be noted, and it should build a list of the people that have already been entered. It should also allow you to prioritize the activities. You should be able to view tasks by date, using the current date first. Priorities would be displayed using different icons or colors. You should also be able to search your tasks. It should include password protection if the smartphone or tablet is lost or stolen.

e. An apartment finder

The apartment finder app should allow you to find apartments based on your current location or select a location, in the case of a move or travel. The app should allow you to select a price range, availability date and other features, such as number of rooms, amenities, etc. It should include a map showing apartments in the result list. When an apartment is selected, it should display the floor plan and perhaps photos of the apartment.

An advanced feature would allow you to select apartments for a short duration, such as travel, an extended business contract or a medical stay.

f. A weather app

Include current date and time, the ability to select a location or use the current location, an icon showing the general weather along with the current conditions, the forecast and the ability to change to show daily, 10 day forecast, travel weather, regional map, etc.

6. *Explore the Joomla! website at www.joomla.org. How could this open source application be helpful in implementing your designs from Problem 4 above? Summarize your findings in a paragraph. Use the Web to find another CMS and use a paragraph to compare it to Joomla! Be sure to address cost, ease of use, support, and availability in your comparison.*

   Joomla! would be useful because it allows you to keep track of every object on your Web page, whether it is text, media, or banners. There is international language support as well. Joomla! allows the developer to create menus, polls, search, and so on. It is easy to use, free, and available as a download.

   There are many other CMS available. Refer to the list in Wikipedia at: http://en.wikipedia.org/wiki/List_of_content_management_systems

7.. *Use brainstorming to develop a new set of widgets (gadgets) to be more productive. Come up with a list of your top five bright ideas for new widgets.*

   The solution will vary considerably from group to group. Some of the widgets that might be useful are weather forecasting, Web cam viewers for various Web cams scattered throughout the park, reminders that notify people about meetings and appointments, and so on.

Central Pacific University—Problems

1. *Use Access to view the HARDWARE INVESTMENT REPORT. If you are familiar with Access, use the* ***File/Export…****menu option to save the report as a Web page. When the* ***Export*** *dialogue box opens, click in the* ***Save As Type*** *dropdown list and select* ***HTML Documents.***

   Report Name: HARDWARE INVESTMENT REPORT. This report is illustrated in Figure E11.1.

2. *Chip, Dot, and Mike participated in several brainstorming sessions resulting in the outlining of several reports. Design (or modify using Access) the HARDWARE MASTER REPORT. This report is large, and you will have to be careful to include all the data in the report area. You may want to have several detail lines for each record. Print the completed report.*

   A page of the Hardware Master Report is illustrated below.

Hardware Master Report

11/06/2012 Page 1 of 20

Hardware Inventory Number	339393
Installed	Yes
Brand Name	Xxxxxxxxxxxxxx
Model	Xxxxxxxxxxxxxxxx
Serial Number	9999999999999999
Campus Description	Central Computer Science
Room Location	99999
Date Purchased	04/15/2013
Purchase Cost	$9,999.00
Replacement	$9,999.00
Memory Size	4
Hard Drive	500
Second Hard	1000
Optical Drive	DVD Read/Write
Warranty	Yes
Maintenance Interval	120
Last Preventive Maintenance	06/23/2010
Number of Repairs	0
Cost of Repairs	$0.00

3. *After meeting with Cher Ware and Hy Perteks to discuss reporting needs, Anna has identified the fields for the partially completed NEW SOFTWARE INSTALLED REPORT. Design (or modify) the report to include the elements found in the data flow repository entry. Is the report a summary or detailed report? In a paragraph, outline the logic that you think the report-producing program must use.*

The report is illustrated below.

Software Category Internet

Title	**Version**	**Operating Number**	**Publisher System**	**Brand /**	**Campus Model**	
Room						
Apple Final Cut Studio	2	Mac OS X Leopard	Apple	Xxxxxxxxxxxxxx Xxxxxxxxxxxxxxxx	Central Administration 99999	1
						1
				Xxxxxxxxxxxxxxxxxx		
				Xxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxx	Central Zoology 22222	1

This report is a detailed report, listing each piece of software received. The logic would include presorting the records by category, title and version, reading a record, locating the campus location in a table file, formatting the output line, and printing the line.

4. *Both Dot and Mike need to know when new computers have been received. Create the NEW COMPUTER RECEIVED REPORT. The COMPUTER RECEIVED REPORT data flow contains the necessary elements.*

The report is illustrated below.

New Computer Received Report

11/06/2012	**Brand Name**	**Model**	**Date Purchased**	**Number of Machines**	Page 1 of 1
	Xxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxx	01/01/2013	3	
	Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	04/15/2013	4	
		Xxxxxxxxxxxxxxxxx	01/01/2013	2	
	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	01/01/2013	3	
		Xxxxxxxxxxxxxxxxxxxx	01/01/2013	11	

5. *Design the SOFTWARE MASTER REPORT containing pertinent information that helps Cher and Hy to locate the various copies of any software package easily. The elements necessary to produce the report are located on the SOFTWARE MASTER REPORT data flow. The TITLE,VERSION, OPERATING SYSTEM NAME, PUBLISHER, CATEGORY, and FIRST and LAST NAME of the software expert should be group printed. Totals are to be included for each TITLE/OPERATING SYSTEM/ VERSION combination. Print the completed report design.*

A portion of the Software Master Report is illustrated below.

Software Master Report

11/06/2012 Page 1 of 7

Title	Adobe Dreamweaver	**Publisher**	Macromedia
Operating System	Mac OS X Leopard	**Category Description**	Internet Design
Version	CS4	**Expert Name**	Amy Rockwell

Inventory Number	**Campus**	**Room**	**Brand Name**	**Hardware Number**	**Site License**	**Number of Copies**
76422076	Central Administration	99999	Xxxxxxxxxxxx	34044449	No	40
76422076	Central Administration	99999	Xxxxxxxxxxxxxx	33453403	No	40
76422076	Central Administration	22222	Xxxxxxxxxxxxxx	10220129	No	40
				Software Subtotal		$594.00

6. *Design the HARDWARE INVENTORY LISTING, showing the software available in each room at each campus. The CAMPUS field should be the CAMPUS DESCRIPTION, not the code representing the campus.*

A portion of the Hardware Inventory Listing is illustrated below:

11/06/2012 **Hardware Inventory Listing** Page 1 of 1

Campus	Room Location	Inventory Number	Brand	Model	Present
Central Administration	11111	84004782	Xxxxxxxxxxx	Xxxxxxxxxxxxxxxxxx	
		90875039	Xxxxxxxxxxx	Xxxxxxxxxxxxxxxx	
		93955411	Xxxxxxxxxxx	Xxxxxxxxxxxxxxxxxx	
		99381373	Xxxxxxxxxxx	Xxxxxxxxxxxxxxxxxx	
		Total number of machines in room		4	

7. *Design the INSTALLED COMPUTER REPORT, showing personal computers that have been installed in each room. Use the CAMPUS DESCRIPTION and group print by CAMPUS DESCRIPTION and ROOM LOCATION. The INSTALLED BOARDS is a repeating group, with up to five entries per computer.*

A portion of the Installed Computer Report is illustrated below:

11/06/2012 Installed Computer Report Page 1 of 2

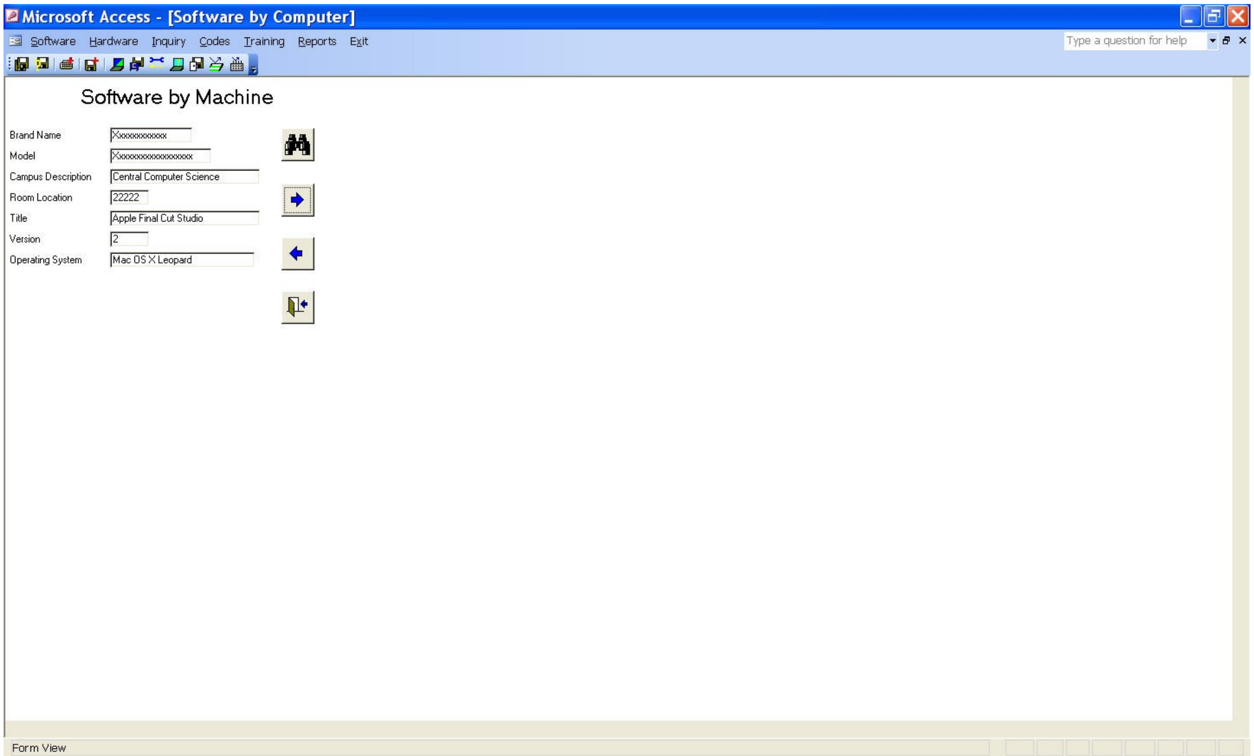
Campus	**Central Administration**			
Room Location 11111	Brand Name	Xxxxxxxxxxxxxxxx	Memory Size	4
	Model	Xxxxxxxxxxxxxxxxxxxx	Hard drive	750
	Inventory Number	84004782	Second Fixed	0
	Optical Drive	Blu-ray		

8. *Use Access to view the SOFTWARE BY CATEGORY screen report. Click the **Find** button and locate **CASE toolset**. Click the **Next** and **Previous** buttons to view next and previous **Software Categories**.*

This display screen is illustrated in Figure E11.2.

9. *Design the SOFTWARE BY MACHINE screen report. Refer to the data flow repository entry for elements.*

The display screen is illustrated below. The Access name for this screen is Software by Computer.



10. *Design the COMPUTER PROBLEM REPORT. This report shows all computers that have a large number of repairs or a large repair cost. Refer to the repository description for the data flow for the elements or modify the Access report.*

The report is illustrated below.

11/06/2012

Computer Problem Report

Page 1 of 1

Inventory Number	Brand Name	Model of	Number Repairs	Cost of Repairs	Warranty
40030303	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	10	$955.25	Yes
34589349	Xxxxxxxxxxxx	Xxxxxxxxxxxxxx	8	$720.00	Yes
56620548	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	8	$487.22	Yes
84004782	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	4	$376.90	Yes
24720952	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	4	$290.00	Yes
93955411	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	6	$242.55	Yes
33453403	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	6	$155.50	Yes
99381373	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	1	$40.00	Yes
10220129	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	2	$20.00	Yes
339393	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
11398423	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
22838234	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
28387465	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
34044449	Xxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
47403948	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
38376910	Xxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
99481102	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
2342	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
1111	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
70722533	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
90875039	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
94842282	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
11111111	Xxxxxxxxxxxxx	Xxxxxxxxxxxxxx	0	$0.00	Yes
			Total cost of repairs	$955.25	
			Number of machines	23	

11. *Design or modify the INSTALLATION REPORT. Refer to the repository entry for the data flow for the elements. This report shows which computers have been recently received and are available for installation.*

The report follows.

11/06/2012

Computer Installation Report

Page 1 of 2

Brand	Model	Inventory Number	Serial Number	Memory	Optical	Hard Drive	Second Drive
Xxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	34044449	9999999999999999	4	DVD	1000	500
Xxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	34589349	9999999999999999	4	DVD Read/Write	1000	750
Xxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	38376910	9999999999999999	6	CD Read/Write	1000	750
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	22838234	9999999999999999	2	DVD Read/Write	1000	0
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	90875039	9999999999999999	4	DVD Read/Write	500	0
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	70722533	9999999999999999	2	Blu-ray Read/Write	2000	1000
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	339393	9999999999999999	4	DVD Read/Write	500	1000
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	10220129	9999999999999999	3	Blu-ray	1000	1000
Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	24720952	9999999999999999	4	Blu-ray	1000	0
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	56620548	9999999999999999	2	DVD	500	0
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	33453403	9999999999999999	8	Blu-ray Read/Write	1000	500
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	47403948	9999999999999999	2	DVD Read/Write	500	750
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	1111	9999999999999999	8	Blu-ray Read/Write	1000	0
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	84004782	9999999999999999	4	Blu-ray	750	0
Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxx	93955411	9999999999999999	4	DVD	500	0

12. *Design the NEW COMPUTER RECEIVED REPORT. Refer to the repository description for the data flow for the elements or modify the Access report. This summary report shows the number of computers of each brand and model. These computers need to be unpacked and have component boards and other hardware installed in them before they may be installed in rooms.*

    This is a duplicate of problem E4—our apologies.

13. *Design or modify the PREVENTIVE MAINTENANCE REPORT. Refer to the repository entry for the data flow for the elements. This report shows which computers need to have preventive maintenance performed on them.*

    A portion of the Preventive Maintenance Report is illustrated below.

11/06/2012 **Preventive Maintenance Report** Page 1 of 1

Week Of 11/6/2009

Campus Location	**Room Location**	**Inventory Number**	**Brand Name**	**Model**	**Last Preventive Maintenance**	**Done Date**
Central Administration	11111	84004782	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxxxx	01/01/2013	
		90875039	Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxx	01/01/2013	
		93955411	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxxxx	01/01/2013	
		99381373	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxxxx	01/01/2013	
	22222	10220129	Xxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxx	06/23/2013	

14. *Design the SOFTWARE CROSS REFERENCE REPORT. Refer to the repository description for the data flow for the elements or modify the Access report. This report shows the computer in which each software package is installed. The TITLE, VERSION, OPERATING SYSTEM MEANING, and PUBLISHER are group printed. The detail lines under the group contain data showing the machine, installation campus, and room.*

    A portion of the Software Cross Reference report is illustrated below.

11/06/2012 **Software Cross Reference Report** Page 1 of 5

Title:	***Adobe Acrobat***		***Publisher:***	***Adobe***
Version:	4		**Operating System:**	Windows XP
Campus	**Room**	**Hardware Number**	**Brand**	**Model**
Central Zoology	22222	28387465	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxxxx
Central Zoology	22222	11398423	Xxxxxxxxxxxxxxxx	Xxxxxxxxxxxxxxxxxxxx
			Total copies of software	2

15. *Design or modify the OUTSTANDING COMPUTER PURCHASE ORDERS REPORT. Refer to the repository entry for the data flow for the elements. This report would be produced for all PURCHASE ORDER records that have a purchase order code of M101, representing computers, with the additional condition that the QUANTITY ORDERED on the record must be greater than the QUANTITY RECEIVED. In a paragraph, state whether this report is a summary, exception, or detailed report. Explain.*

The Outstanding Computer Purchase Orders report is illustrated below.

11/06/2012 **Outstanding Computer Purchase Orders** Page 1 of 1

Order Number	Date Ordered	Brand Name	Model	Quantity Ordered	Quantity Received
20322	02/03/2012	Xxxxxxxxxxxxxxx	Yyyyyyyyyyyy	20	20
20345	03/21/2012	Aaaaaaaaaaaa	Bbbbbbbbbbbb	10	10
20376	04/15/2012	Mmmmmmmmm	Nnnnnnnnnnnn	9	9
20384	04/15/2012	Rrrrrrrrrrrrrrr	Sssssssssssss	20	10
20399	06/14/2012	Wwwwwwwwww	Zzzzzzzzzzzzzz	18	10
20412	06/30/2012	Ggggggggggg	Hhhhhhhhhhhhh	22	12

This is an exception report. The report is produced by selecting a subset of records from the Purchase Order file. The first selection criteria is that records must be in the Computer category (the purchase order code is M01) and the second category is that the records must represent outstanding purchase orders (the quantity received is less than the quantity ordered).

16. *Design the SOFTWARE INVESTMENT REPORT. Refer to the repository description for the data flow for the elements or modify the Access report.*

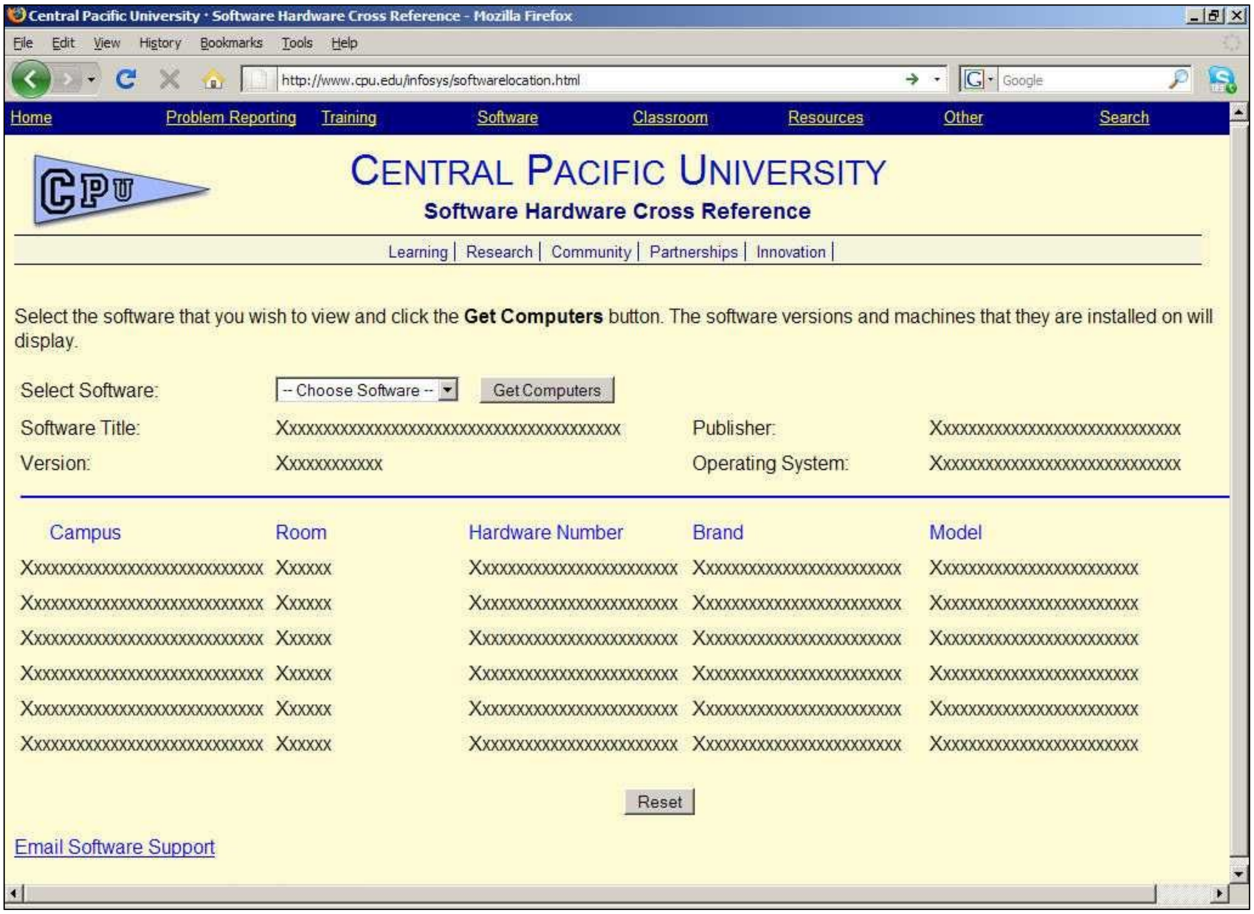
A portion of the Software Investment report is illustrated below.

11/06/2012 **Software Investment Report** Page 1 of 2

Title	Version	Operating System	Site	Number of Copies	Unit Cost	Software Cost
Acrobat	6	Windows XP	No	10	$590.00	$5,900.00
Adobe Acrobat	4	Windows XP	No	12	$100.00	$1,200.00
Adobe DreamWeaver	CS4	Windows XP	No	20	$299.00	$5,980.00
Adobe DreamWeaver	CS4	Mac OS X	No	40	$198.00	$7,920.00
Adobe Flash	CS4	Windows XP	No	40	$35.00	$1,400.00
Adobe Photoshop	CS4	Windows XP	No	20	$289.00	$5,780.00
Adobe Premiere	7	Windows Vista	No	20	$120.00	$2,400.00
After Effects	4.1	Mac OS X	No	20	$259.00	$5,180.00
Apple Final Cut	2	Mac OS X	Yes		$1,299.00	$1,299.00

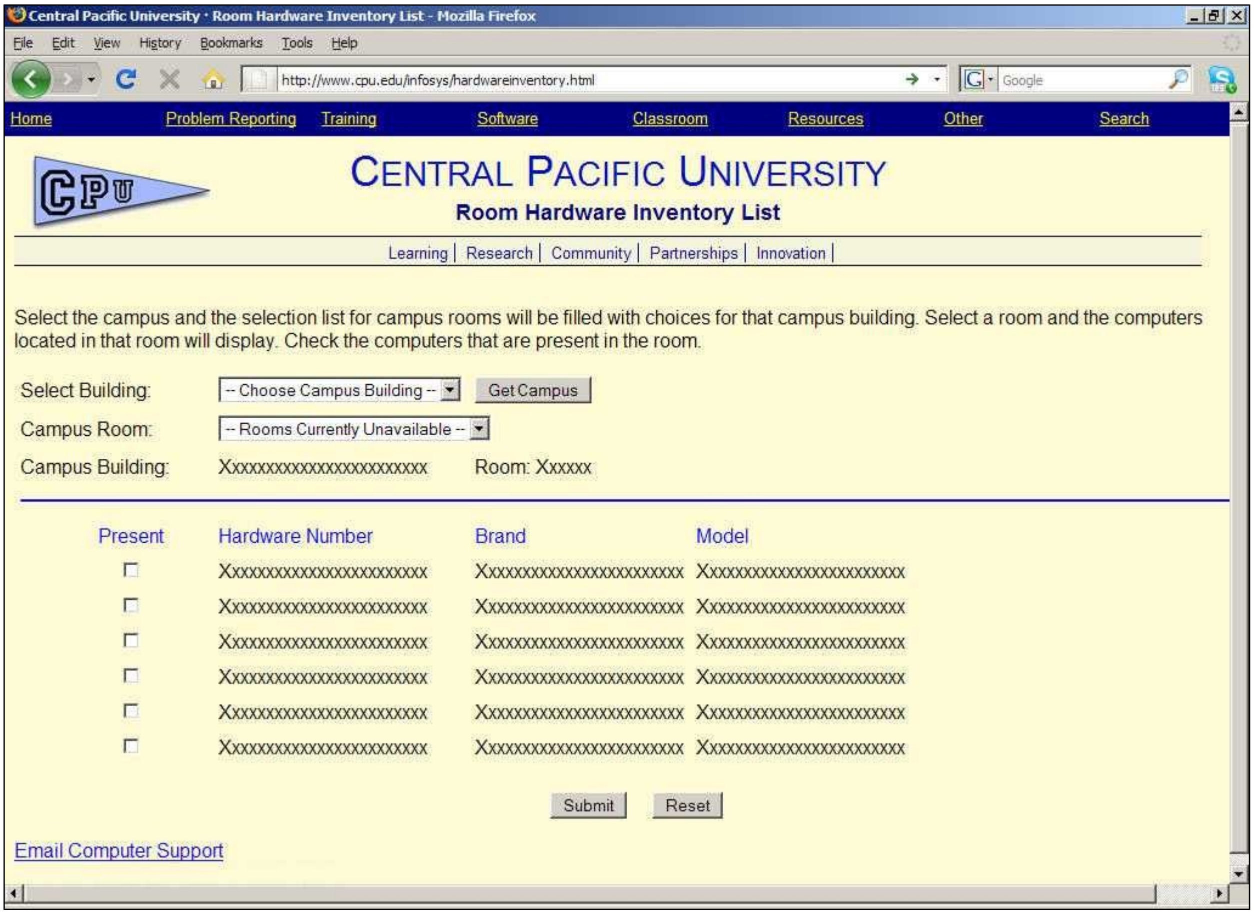
17. Design the SOFTWARE CROSS REFERENCE Web page. Refer to the repository description for the SOFTWARE CROSS REFERENCE REPORT data flow for the elements on the Web page. This Web page shows the computers that each software package is installed. Include a drop-down list of software that allows the user to select a software package. The design uses Ajax to refresh the Web page list of computers containing the software and their locations.

The Web page is illustrated below.



18. *Design the HARDWARE INVENTORY LISTING Web page, showing the computers available in each room at each campus. The CAMPUS is selected from a drop-down list displaying the CAMPUS DESCRIPTION. When the user selects a campus name from the drop-down list, the Web page uses Ajax techniques to fill the campus room drop-down list. When a room is selected, the Web page uses Ajax to display the machines located in the room. Use the repository for the HARDWARE INVENTORY LISTING without total number of machines at campus or the total number of machines.*

The Web page is illustrated below.


Central Pacific University - Room Hardware Inventory List - Mozilla Firefox
File Edit View History Bookmarks Tools Help
http://www.cpu.edu/infosys/hardwareinventory.html
Google
Home Problem Reporting Training Software Classroom Resources Other Search
CPU
CENTRAL PACIFIC UNIVERSITY
Room Hardware Inventory List
Learning | Research | Community | Partnerships | Innovation |
Select the campus and the selection list for campus rooms will be filled with choices for that campus building. Select a room and the computers located in that room will display. Check the computers that are present in the room.
Select Building: -- Choose Campus Building -- Get Campus
Campus Room: -- Rooms Currently Unavailable --
Campus Building: Xxxxxxxxxxxxxxxxxxxxxxxx Room: Xxxxxx
Present Hardware Number Brand Model
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx Xxxxxxxxxxxxxxxxxxxxxxxx
Submit Reset
Email Computer Support